



Lab: Prepare a Data Model for a Transaction Processing System

Estimated time: 30 minutes

Learning objectives

After completing this lab, you will be able to:

- Apply data modeling principles by designing a normalized data model for a retail bank's transaction processing system.
- Create an Entity Relationship Diagram (ERD) in Lucidchart to visually represent the entities, attributes, and relationships in the retail bank's transaction processing system.

Prerequisites:

You can complete your tasks using any text editor, such as Word, Google Docs, or Notepad.

- If you install Microsoft Word, you can proceed directly with the instructions. However, if you don't have Microsoft Word, you can sign up for a free version of Microsoft 365 online. Follow [these instructions](#) to create your account.
- You can also use Word Online with a Microsoft 365 subscription or access through a free account. If you're new to Word and would like to learn how to get started, you can find helpful resources here: [Getting Started with Word Online](#).
- For this lab, you can use [Lucidchart](#). You only need the free version of LucidChart. If you choose to purchase a paid plan, you will be responsible for any associated charges. If you've not yet signed up for the free account, you can follow the [instructions](#) to get started. You may alternatively use applications like [Microsoft Visio](#).

Introduction

A data model is a structured frame that defines, organizes, and manages data, forming the fundamental blueprint for effectively designing databases. It ensures consistency, usability, and effective data management by combining models into conceptual, logical, and

physical types-specific models dealing with various levels of abstraction targeted toward different stakeholders.

The data model is very important in transaction processing systems like banking. It facilitates account management and ensures data integrity while supporting the smooth operation of deposits, withdrawals, and balance maintenance, all of which form the backbone of reliable and efficient data processing systems.

Exercise: Prepare a data model for a transaction processing system of a retail bank using Lucidchart

In this lab, you will design a comprehensive data model for a retail bank's transaction processing system. The system manages key operations such as account opening, balance maintenance, fixed deposits (FD), recurring deposits (RD), deposits, withdrawals, and savings/checking account types. You will define entities, their attributes, and relationships through step-by-step guidance, culminating in creating an Entity Relationship Diagram (ERD) using Lucidchart. This visual representation will provide a clear and organized view of the data structure, ensuring consistency and usability for seamless transaction processing.

Scenario

A robust data model is required by a retail bank to efficiently manage its transaction processing system in operations like deposits, withdrawals, transfers, and loan repayments.

It must offer ample customer profiles with multiple account types, such as savings, checking, and loans, and detailed transaction logs with timestamps and traceability.

Some of the key requirements would include ACID compliance, multicurrency transaction handling, and role-based access controls to ensure security.

Furthermore, the system needs to embed real-time fraud detection and regulatory compliance. For its sustainability and future capability, the model must incorporate advanced reporting and analytics to trace the transaction pattern with the efficiency of operation, integrity of data, and customer experience over the services delivered in the banking industry.

Let's begin!

Template to use

If you are using Microsoft Word, right-click the link and download the [template](#) document. You can use the template to perform the tasks outlined in the lab.

You can also use the [PDF format](#) for any other word-processing software.

Task 1: Understand the requirements

Step 1: Clearly outline the system's primary features and capabilities and specify the data requirements necessary for the system to execute each operation effectively.

Example

Primary features of the system
Account opening and closure

Step 2: Identify the necessary data points for each operation.

Example

Necessary data points
Account holder details

Task 2: Identify key entities

Clearly determine and define the primary components or building blocks of the system.

Example

Key entities	Description	Key attributes
Customer	Stores customer details	Name, contact, address, ID proof

Task 3: Define attributes for each entity

Identify and specify the key characteristics or properties that describe each entity.

Example

Key entities	Attributes and data types
Customer	• Customer_ID (Primary Key) • Name (VARCHAR) • Address (VARCHAR) • Contact_Info (VARCHAR) • ID_Proof (VARCHAR)

Task 4: Establish relationship between entities

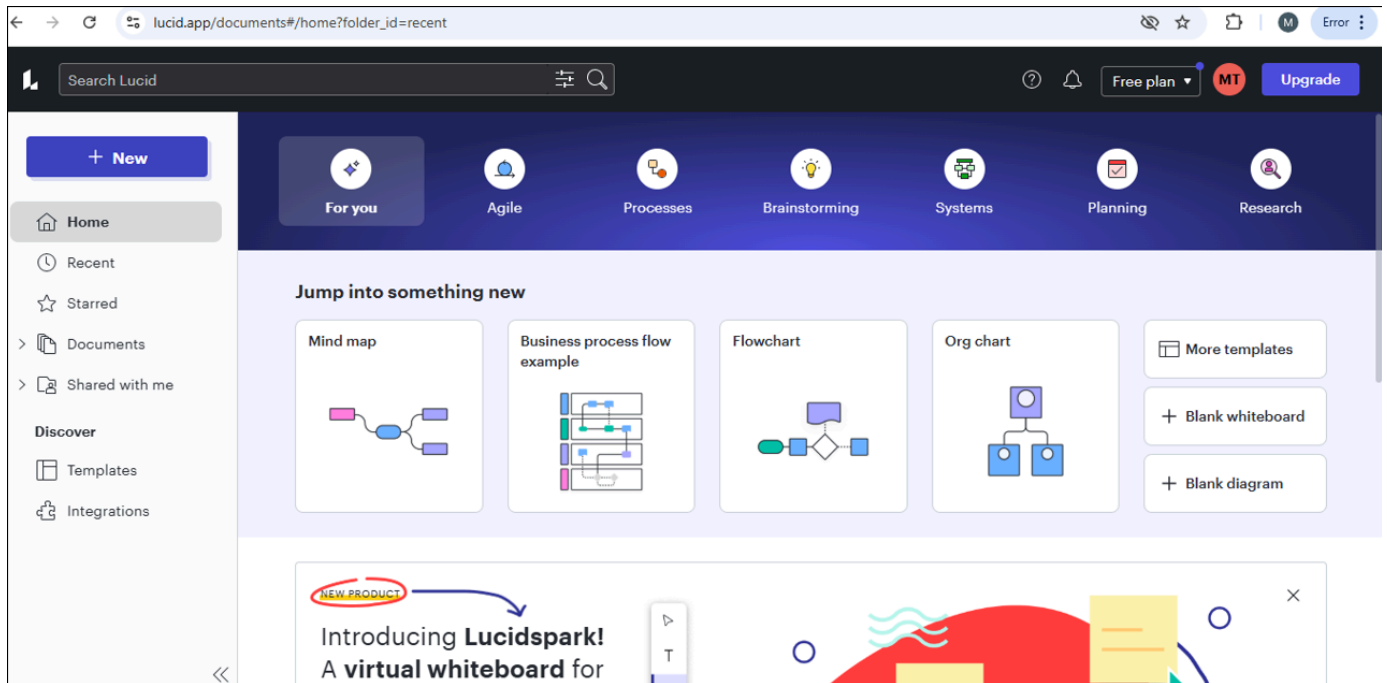
Clearly define how the entities within the system are connected or interact with one another.

Example

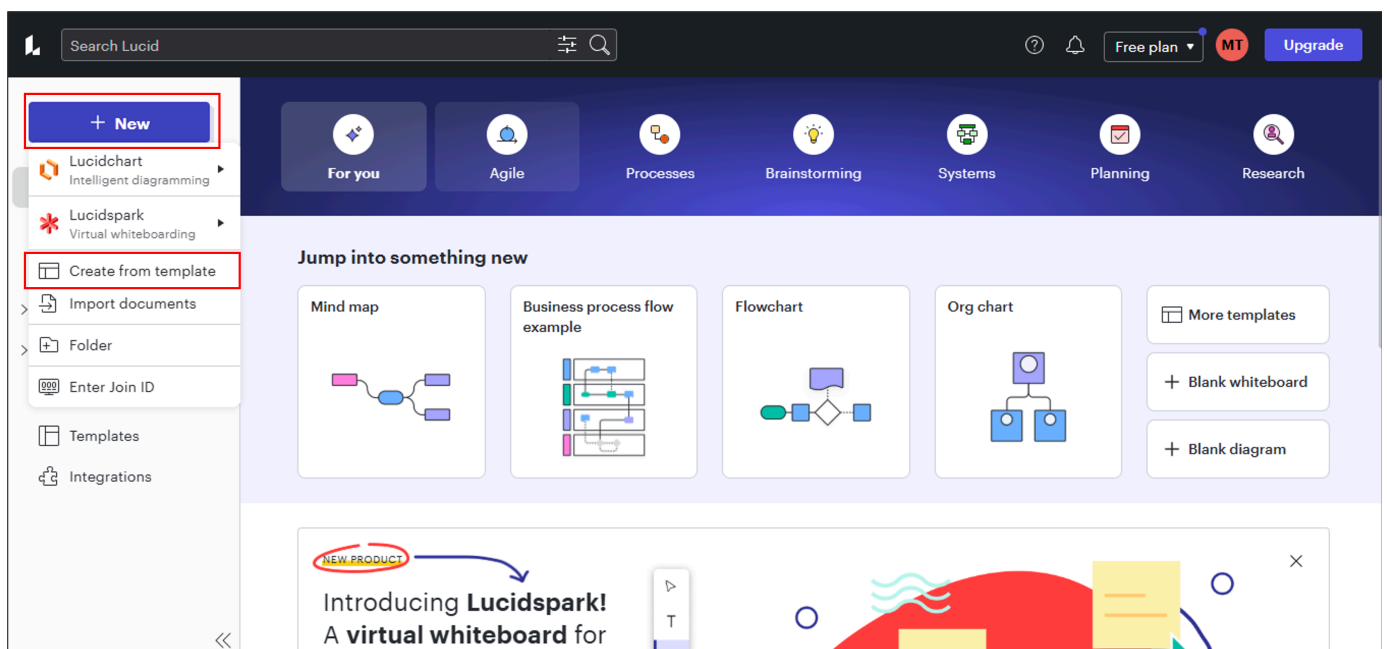
Entities	Relationships	Cardinality
Customer	can have multiple	1:N

Task 5: Create the ERD using Lucidchart

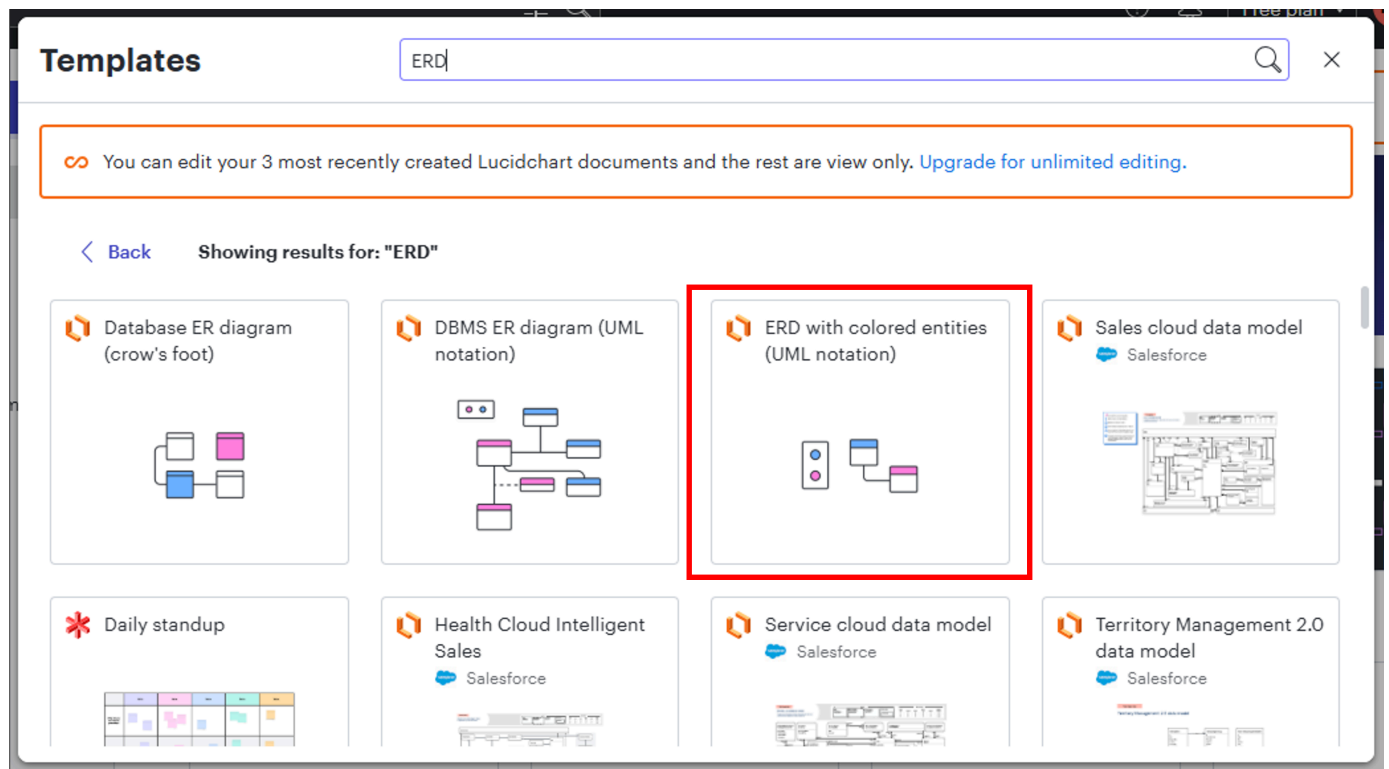
Step 1: Open [Lucidchart](https://lucidchart.com) and login.



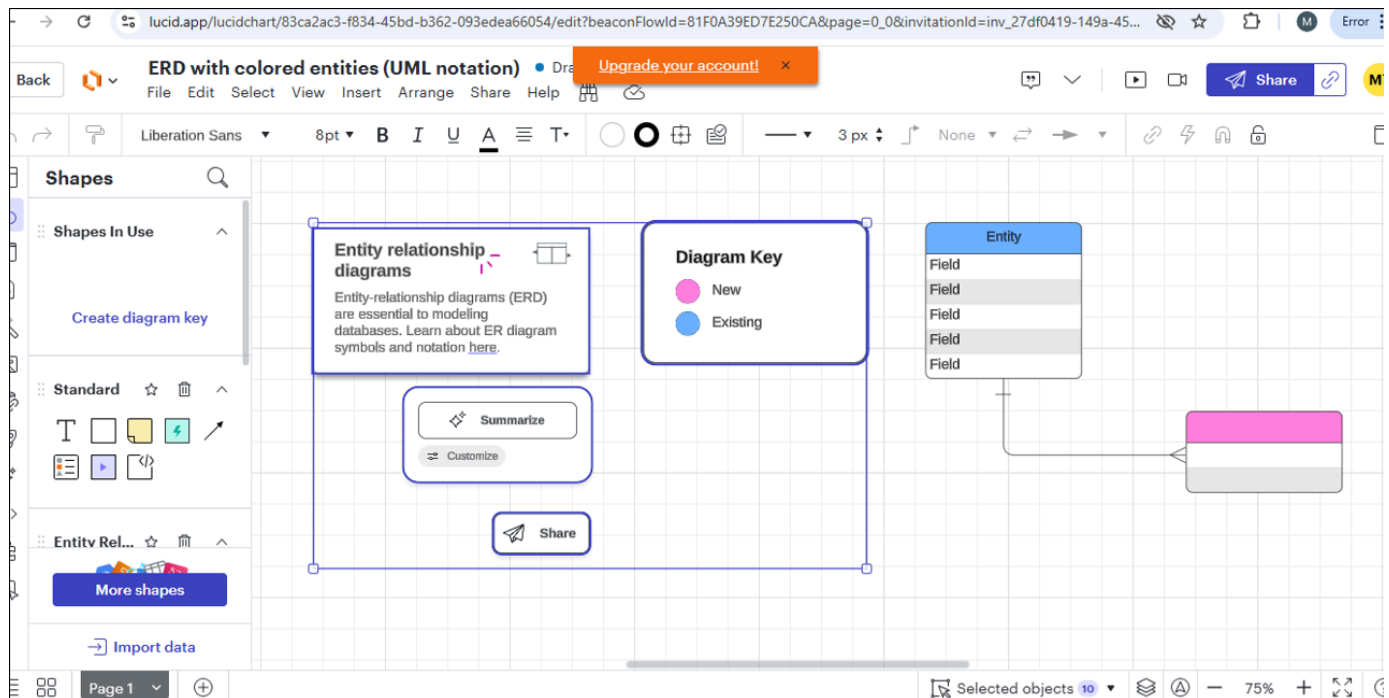
Step 2: Tap on **+New**, and then hit the **Create from template** button.



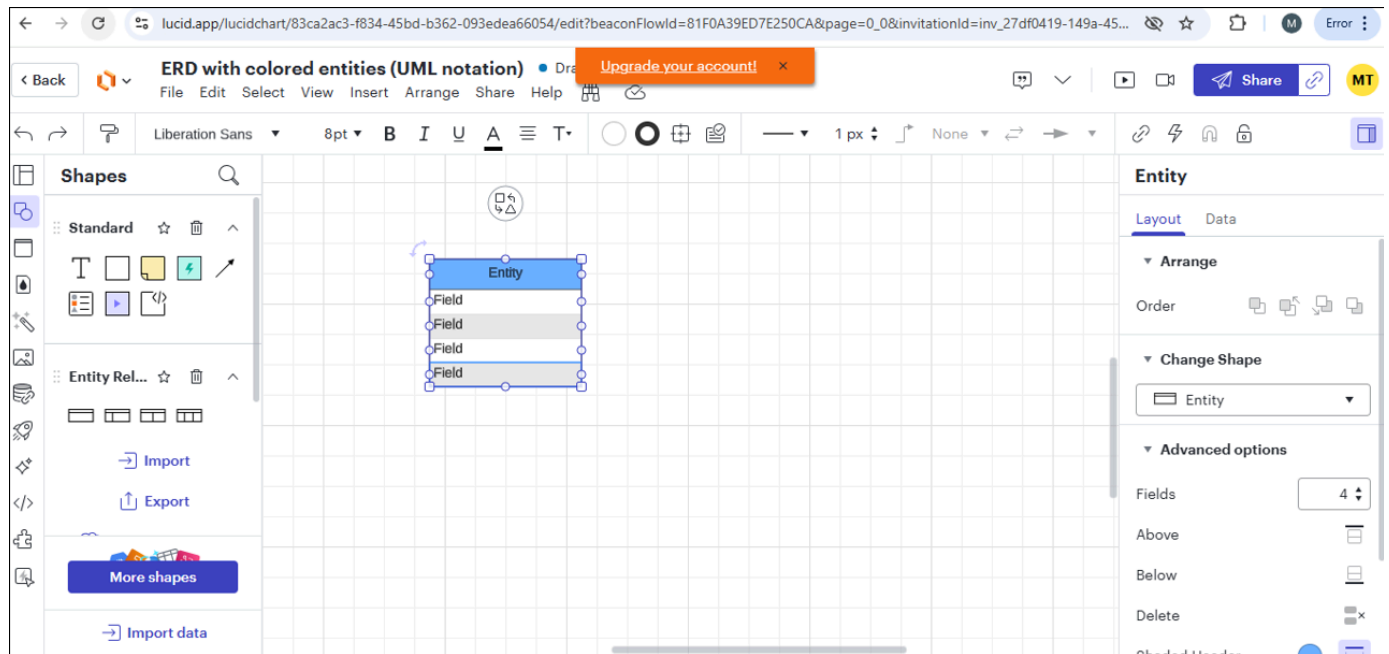
Step 3: Search and choose the ERD template and start using it.



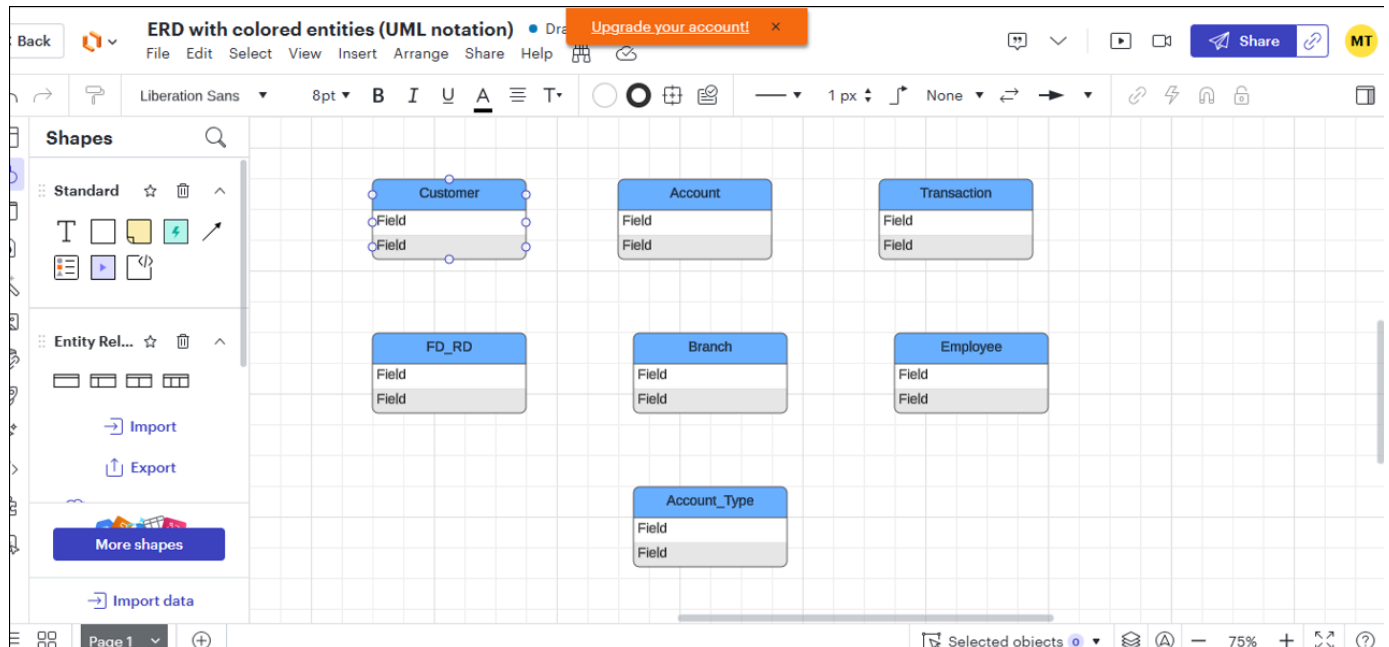
Step 4: Delete all unnecessary items on the board, except for the entity box. To keep it simple, you can also remove extra fields within each entity box.



The rectangle shape will be used to represent each entity.

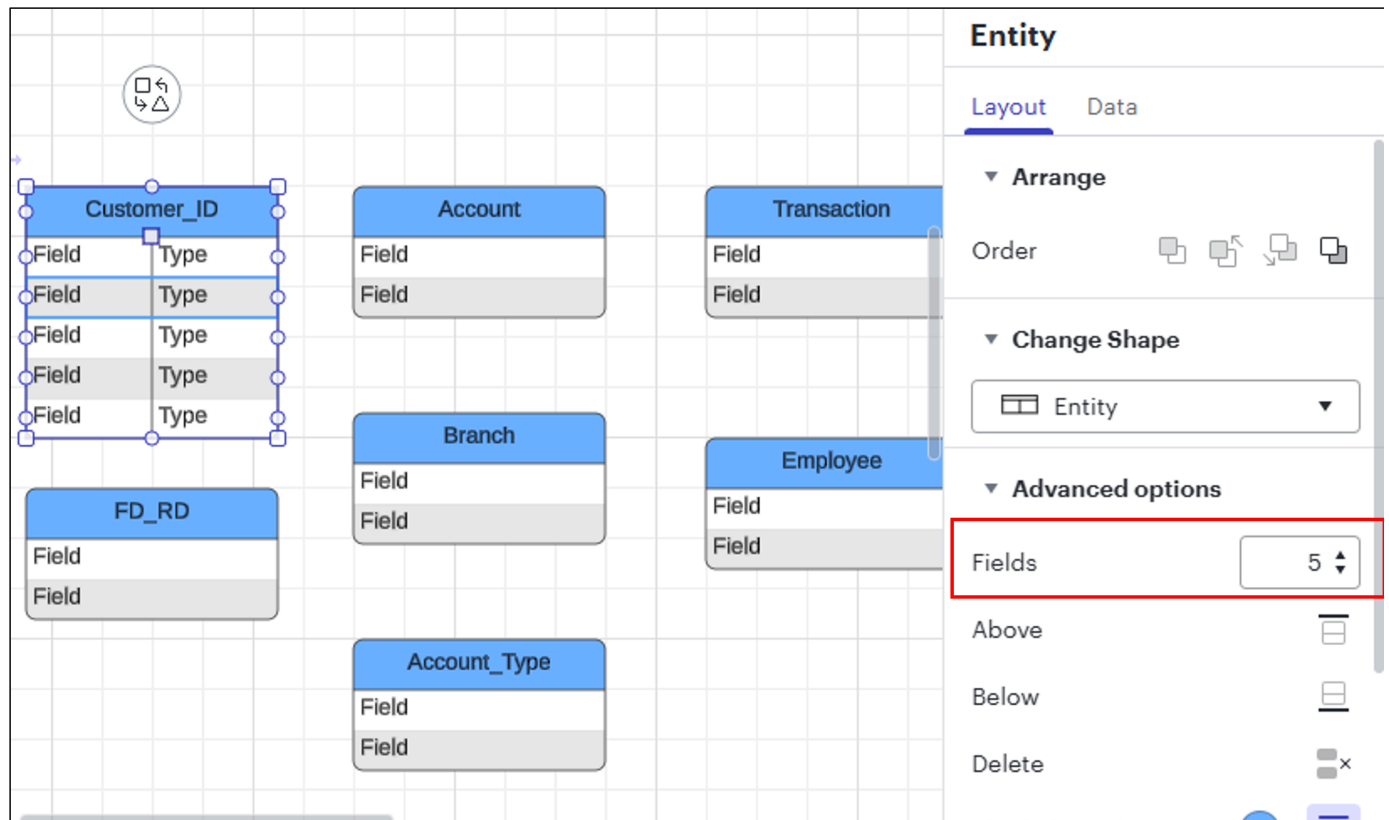


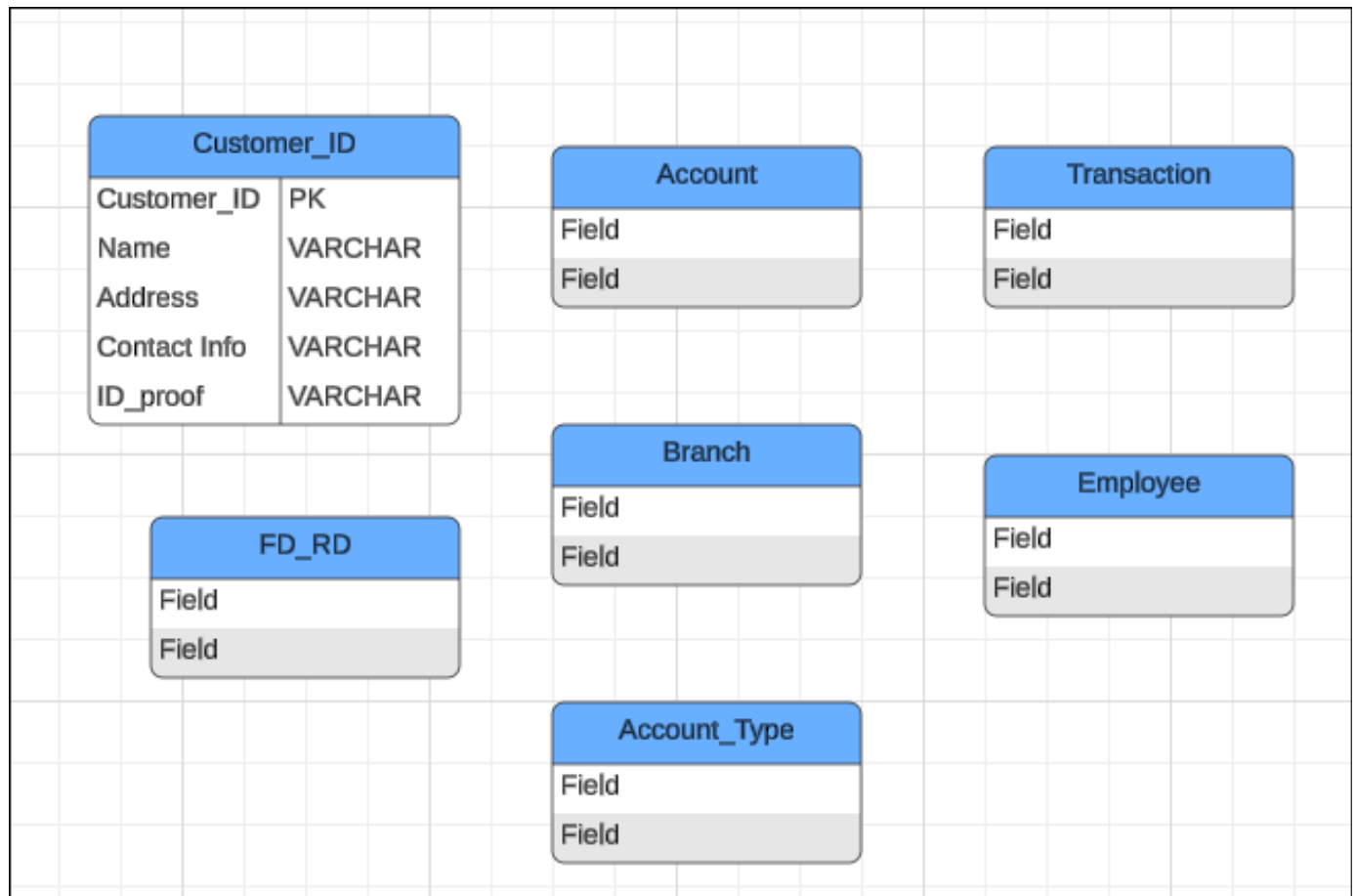
Step 5: Replicate as many entity boxes as you have listed in the table and name them all.



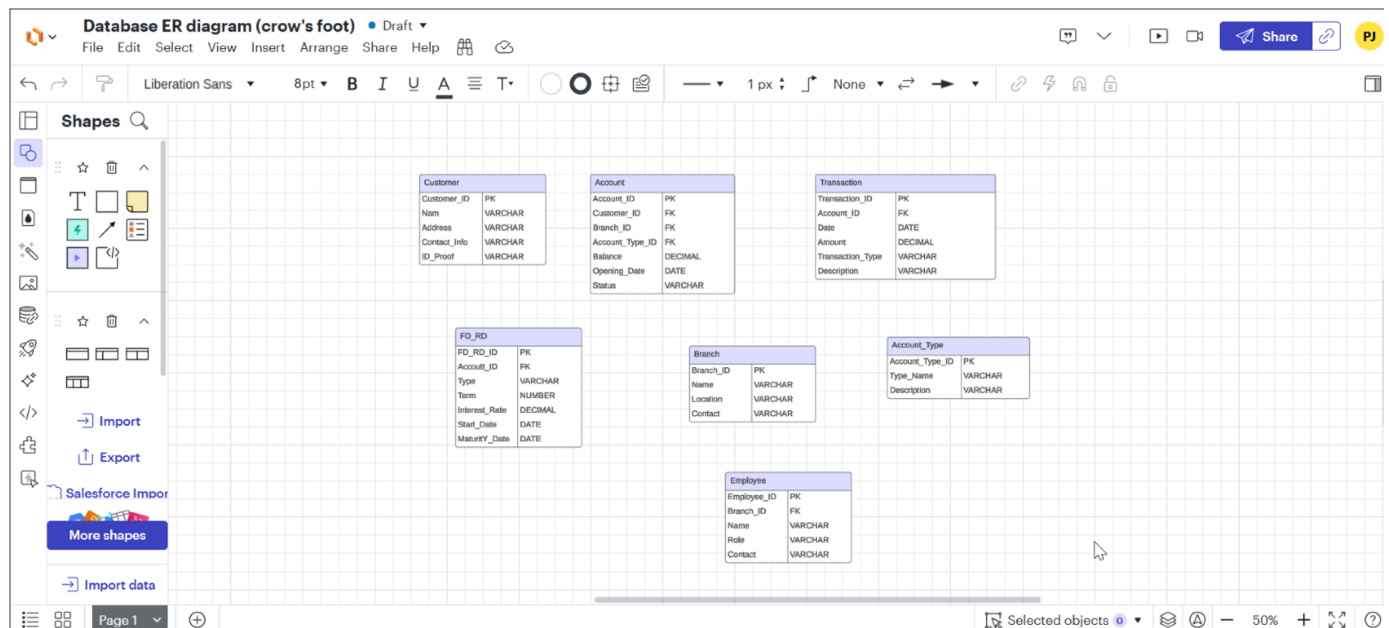
Step 6: List attributes: To list the key attributes, first select a rectangle and choose an entity shape. Then, specify the required number of fields and start filling in the details.

Write the attributes (key fields) inside the rectangle beneath the entity name along with their datatypes. Ensure the attributes are clear and directly associated with their respective entities for easy identification.

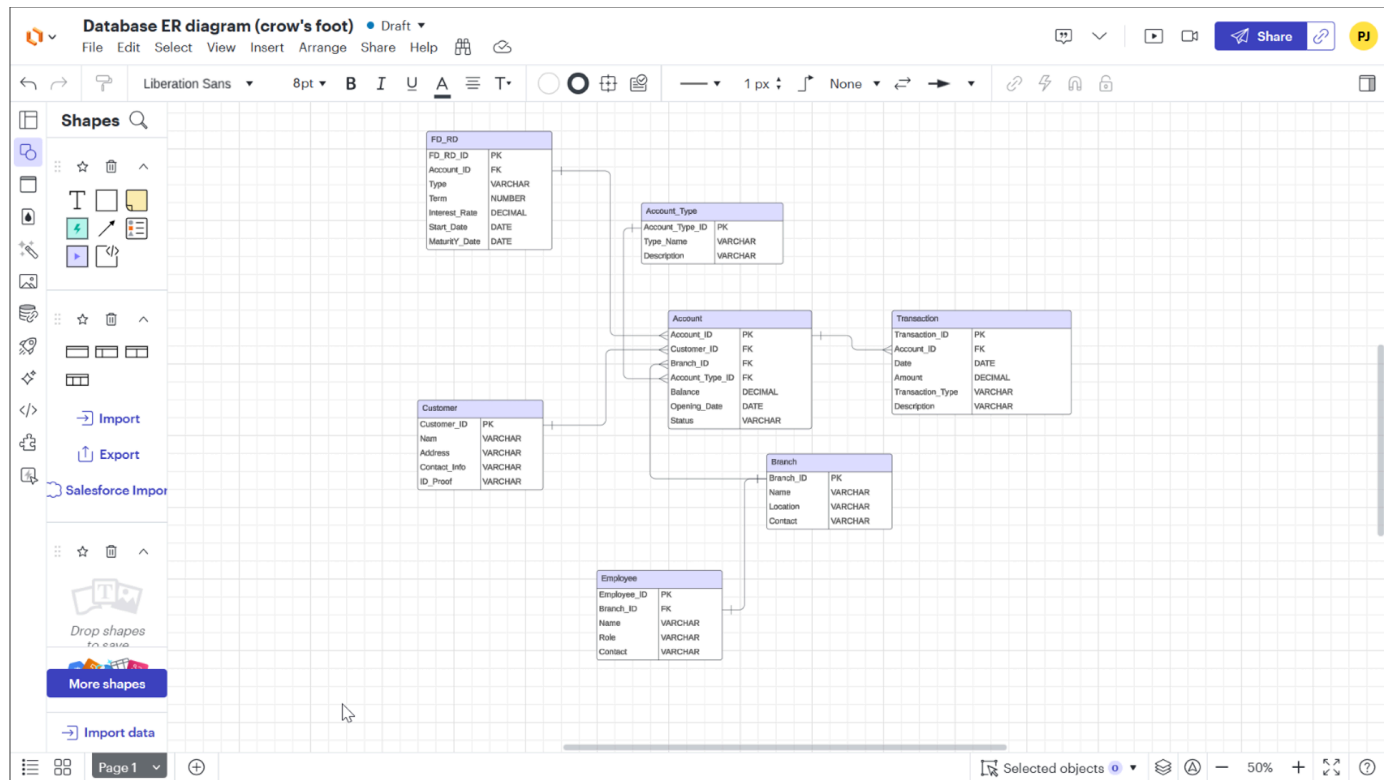




Similarly, list the key attributes for all the entities.



Step 7: Use **lines** to connect entities based on relationships. Add **crow's foot notation** to represent the cardinality (**1:N, 1:1, N:N**).



Summary

Congratulations on completing the hands-on lab, **Prepare a Data Model for a Transaction Processing System**. In this lab, you explored how to apply data modeling principles by designing a normalized data model and creating an Entity Relationship Diagram (ERD) in Lucidchart to visually represent the entities, attributes, and relationships in a processing system.

You can right-click to refer to the [exemplar](#) document to create an ERD for visually representing the entities, attributes, and relationships in a processing system.

Please remember that the exemplar is just a point of view and not a definitive solution.